

Midterm Exam

03/03/2026

- (1) Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest **m-path** between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path and show the path on the figure below if the path exists. (15 pts)

Solution:

The shortest m-path between p and q has the length of 6.

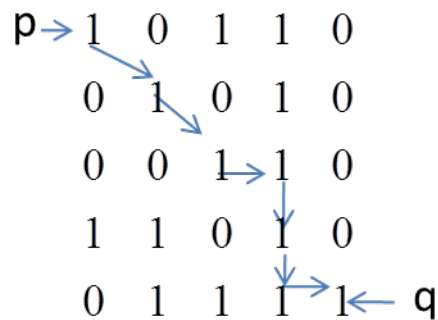


Figure 1- Solution: The shortest m-path between p and q is marked with the length of 6

- (2) To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain. (15 pts)

Solution:

First, it needs a longer exposure time to capture an image under low illumination, which may result in motion blur for the fireworks due to moving fireworks and camera shake. An image deblurring process is needed.

Second, an image taken with poor illumination usually contains high level noise. Hence, it is need to reduce the noise. Assuming the noise is zero-mean Gaussian, mean filters can be employed. An optional image sharpening operation can be performed after image smoothing.

Finally, an image taken under low illumination usually has a small contrast ratio. However, the objects of interest are fireworks, which have high intensity. You don't want to stretch the contrast for the dark background. Instead, you want to show more details of firework. A histogram specification or an inverse-log function can be used on the intensity component of HIS model.

1. A, B, C, and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set. (20 pts)

Solution:

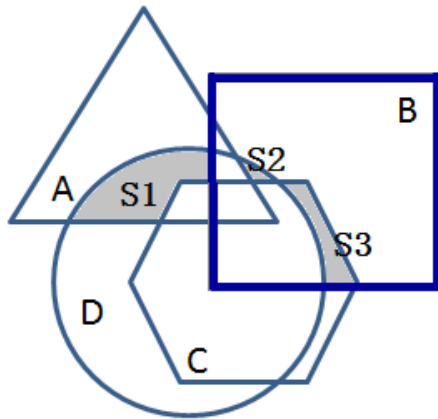


Figure 2- Solution. A graph showing 4 sets: A, B, C, D, with shaded areas marked.

$$S_1 = A \cap D - B - C$$

$$S_2 = B \cap D - C - A$$

$$S_3 = C \cap B - D$$

The set is defined as

$$S_1 \cup S_2 \cup S_3$$

2. Given a 3x3 spatial filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$, answer the following questions:

(a) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? **Why?** (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?) (10pts)

(b) Perform the image **convolution** using **this filter** on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$. Give the

CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping.) (15pts)

Solution:

(a) This mask is an image sharpening filter based on unsharp filtering.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) = \frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

(b) The cropped filtering result is

$$\begin{bmatrix} 2.3333 & 3.7778 & 18.8889 \\ 8.7778 & 26.5556 & 14.2222 \\ 5.6667 & 10.7778 & 24.1111 \end{bmatrix}..$$

3. An image with intensities in the range $r \in [0, L - 1]$ has the pdf $p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L - 1 \\ 0 & \text{otherwise} \end{cases}$, where $L-1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L - 1 \\ 0 & \text{otherwise} \end{cases}$. Assume that the intensity value is **continuous** and find the transformation function $z=T(r)$ in terms of r and z that will accomplish this. (25 pts)
- (Hints: the transformation function of histogram equalization is given as $s = T(r) = (L - 1) \int_0^r p_r(w)dw$.)

Solution:

$$s = T(r) = (L - 1) \int_0^r p_r(w)dw = (L - 1) \int_0^r \frac{4w}{3(L-1)^2} dw = \frac{2r^2}{3(L-1)}$$

$$s = G(z) = (L - 1) \int_0^z p_z(w)dw = (L - 1) \int_0^z \frac{2w}{(L-1)^2} dw = \frac{z^2}{(L-1)}$$

Since $s = T(r) = G(z)$, we have

$$\frac{z^2}{(L-1)} = \frac{2r^2}{3(L-1)}$$

Then

$$z_{1,2} = \pm \sqrt{\frac{2r^2}{3}}$$

Since z is non-negative,

$$z = \sqrt{\frac{2r^2}{3}}$$

Bonus question

Perform the Fourier transform to a function as shown in Figure 3. (20 bonus points)

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt$)

Solution:

The key point to solve this problem is to write $f(t)$ as a summation of three individual functions such that

$$f(t) = \begin{cases} 3 & -2 \leq t \leq -1 \\ 2 & -1 < t \leq 2 \\ 3 & 2 < t \leq 4 \\ 0 & \text{otherwise} \end{cases} = f_1(t) + f_2(t) + f_3(t)$$

Where

$$\begin{aligned} f_1(t) &= \begin{cases} 3 & -2 \leq t \leq -1 \\ 0 & \text{otherwise} \end{cases} \\ f_2(t) &= \begin{cases} 2 & -1 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases} \\ f_3(t) &= \begin{cases} 3 & 2 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Therefore, based on the linearity property of Fourier transform, we have

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt = \int_{-\infty}^{\infty} f_1(t) e^{-j2\pi ut} dt + \int_{-\infty}^{\infty} f_2(t) e^{-j2\pi ut} dt + \int_{-\infty}^{\infty} f_3(t) e^{-j2\pi ut} dt \\ &= \int_{-2}^{-1} 3 e^{-j2\pi ut} dt + \int_{-1}^2 2 e^{-j2\pi ut} dt + \int_2^4 3 e^{-j2\pi ut} dt \\ &= \frac{3}{-j2\pi u} (e^{j2\pi u} - e^{j4\pi u}) + \frac{2}{-j2\pi u} (e^{-j4\pi u} - e^{j2\pi u}) + \frac{3}{-j2\pi u} (e^{-j8\pi u} - e^{-j4\pi u}) \\ &= 3 * \left[\frac{\sin(\pi u)}{\pi u} \right] e^{j3\pi u} + 6 * \left[\frac{\sin(3\pi u)}{3\pi u} \right] e^{-j\pi u} + 6 * \left[\frac{\sin(2\pi u)}{2\pi u} \right] e^{-j6\pi u} \end{aligned}$$